Module-3

# Data Exploration using Inferential Statistics

- Overview of Various forms of distributions: Normal, Poisson Distribution (Written Notes: uploaded separately)
- Central limit theorem
- Confidence Interval
- Test Hypothesis: Z-test, t-test, Type-I, Type-II Errors
- ANOVA

# Central Limit Theorem

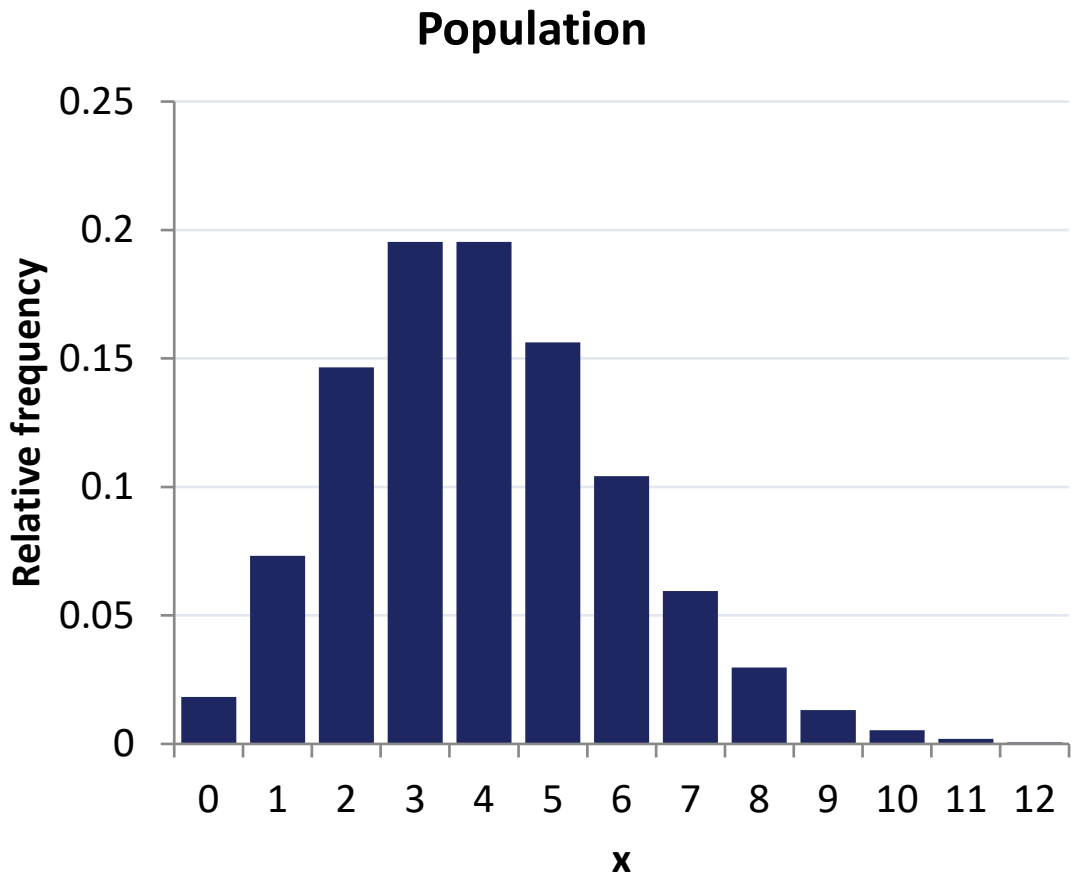
If you take many random samples from *any* population and calculate their averages, those averages will form a normal, bell-shaped distribution — regardless of the population's original shape.

*Holds true once sample size is reasonably large — the common rule of thumb is $n \geq 30$.*

# Example: A Poisson Population

## 1 Start with a skewed population

The values follow a Poisson distribution— not symmetric, not bell-shaped.



# Example: A Poisson Population

1. **Start with a skewed population**

   The values follow a Poisson distribution (mean = 4) — not symmetric, not bell-shaped.

2. **Draw a sample of 50**

   Take 50 random values from the population and calculate their average.

3. **Repeat 10,000 times**

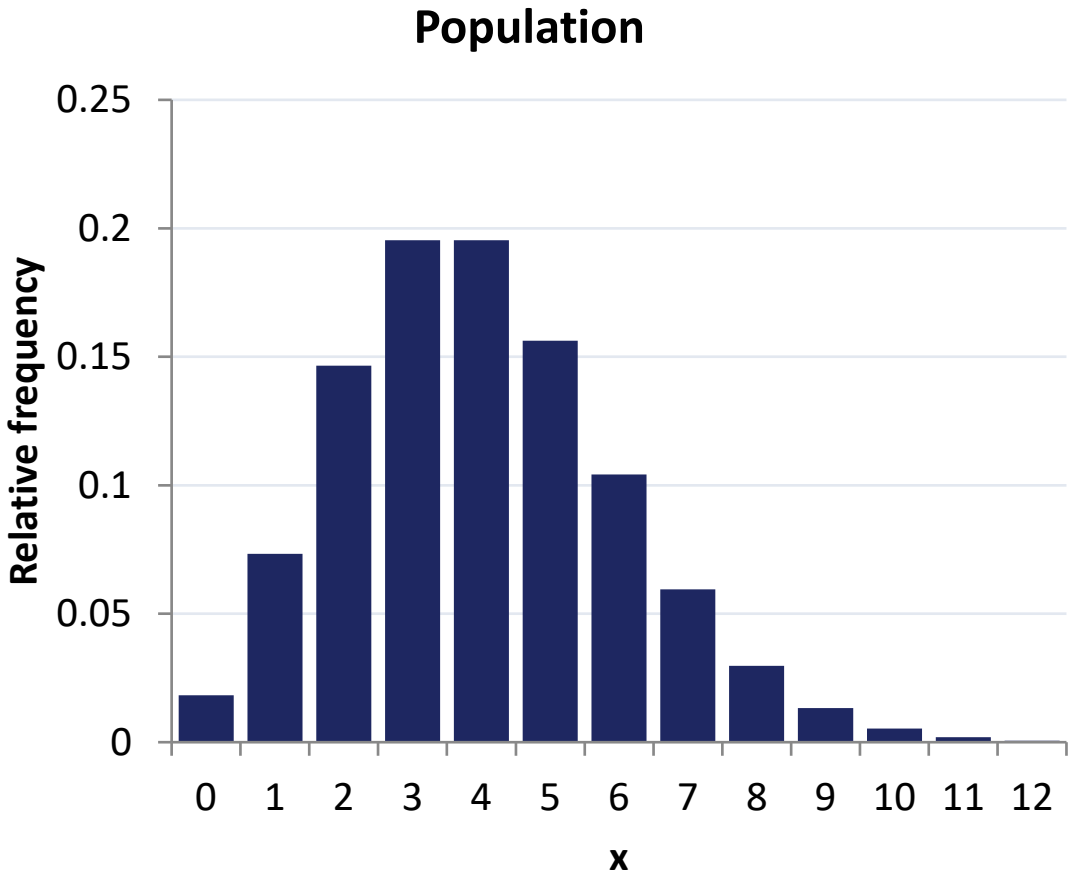
   Draw a new sample of 50 each time, and record each sample's average.

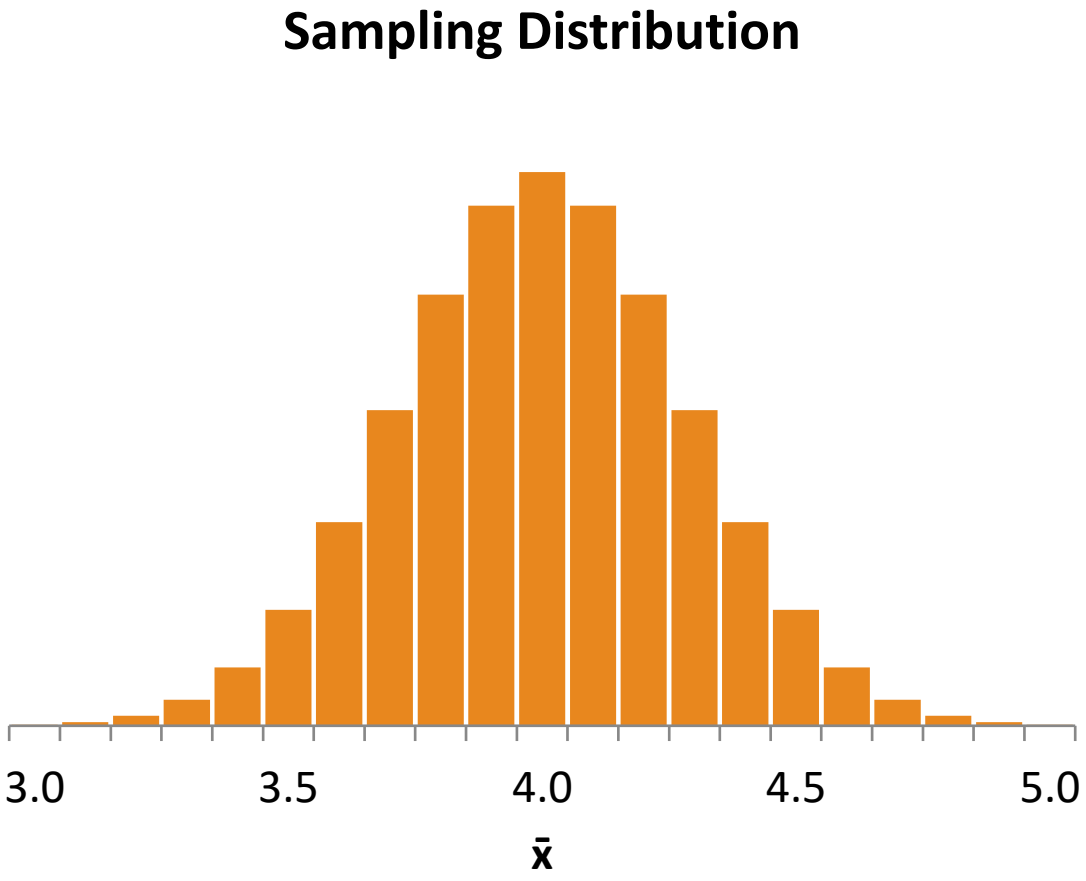
4. **Plot the 10,000 averages**

   The resulting histogram of sample means is smooth, symmetric, and normal.

# Example: A Poisson Population

Population follows a Poisson distribution. Draw 10,000 samples, each of size n = 50, and average each one.



# A Uniform (Dice) Example

*Population: one fair die roll → 1,000 samples of size n = 30, each averaged*

**1** **Start with a uniform population**

Each face 1–6 equally likely — a flat distribution.

**2** **Draw a sample of 30**

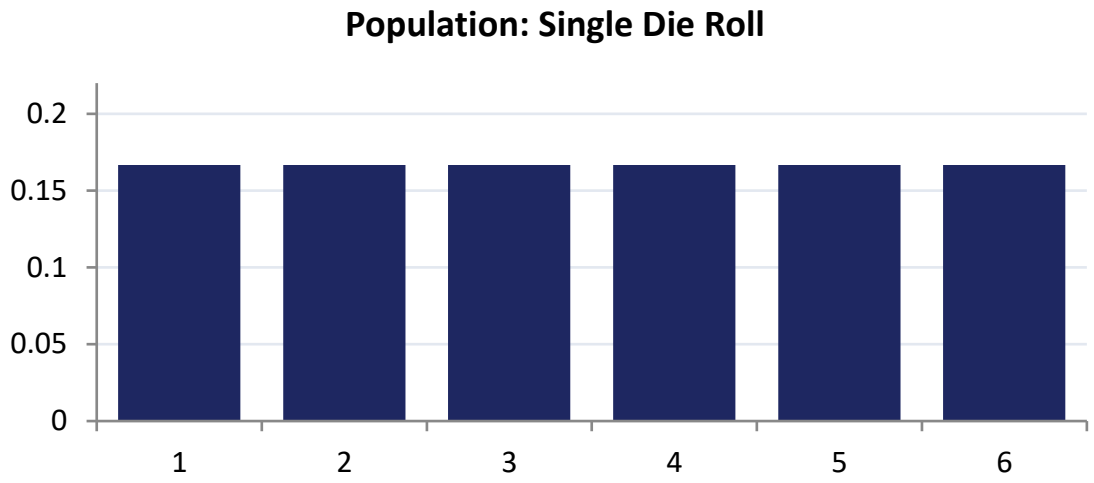
Roll the die 30 times, average the rolls.

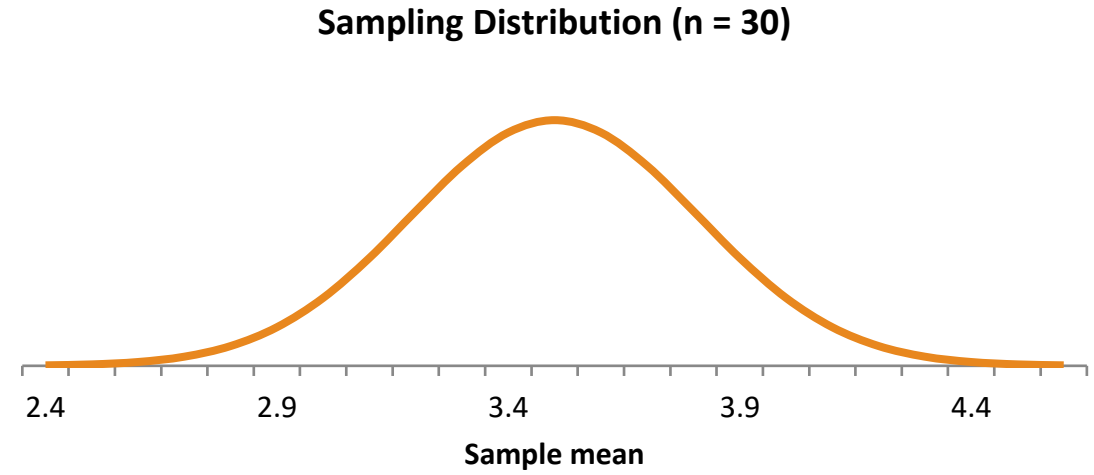
**3** **Repeat 1,000 times**

Roll 30 more each time; record each average.

**4** **Plot the averages**

The histogram of sample means turns normal.



# The Theorem: Standardized Sample Mean

If $\bar{x}$ is the mean of a random sample of size n, drawn from a population with mean μ and standard deviation σ, then — provided the sample size n is large (n ≥ 30) — the sampling distribution of the sample mean $\bar{x}$ approximates a normal distribution with mean μ and standard error σ/√n. This is established by the Central Limit Theorem, stated below without proof.

**Theorem**

If $\bar{x}$ is the mean of a sample of size n drawn from a population with mean μ (Mean of population) and standard deviation σ, then the standardized sample mean

$$z = (\bar{x} - \mu) / (\sigma/\sqrt{n})$$

*is a random variable whose distribution function approaches that of the standard normal distribution as n → ∞.*

# Confidence Interval Estimation

A method for estimating an unknown population parameter with a **range of possible values**, along with a stated level of confidence that the range contains the true value.

**Point estimate**

"The average is 170 cm." — a single number

**Interval estimate**

"We're 95% confident it's between 168–172 cm." — a range plus a confidence level.

**Built on CLT**

The margin of error uses the same standard error (σ/√n) from the Central Limit Theorem.

## Formula

$$CI = \bar{x} \pm z \cdot (\sigma/\sqrt{n})$$

| | |
|---|---|
| $\bar{x}$ | sample mean (point estimate) |
| $z$ | critical value for the confidence level |
| $\sigma/\sqrt{n}$ | standard error of the mean |

# Confidence Interval

### Common z-Values

| | |
|---|---|
| 90% confidence | $z = 1.645$ |
| 95% confidence | $z = 1.96$ |
| 99% confidence | $z = 2.576$ |

### What Widens or Narrows the Interval

**↑ Higher confidence level**

Wider interval (e.g., 99% vs 90%)

**↑ Larger sample size (n)**

Narrower interval

**↑ More population variability (σ)**

Wider interval

## When to Use t Instead of z

If the population standard deviation (σ) is unknown— and the sample size is small ($n < 30$), use the t-distribution instead of the normal distribution, with degrees of freedom = $n - 1$.

# Confidence Interval: Example

*Sample of 50 customers, 95% confidence level*

**1** **Gather the sample statistics**

$\bar{x} = 45$ min, $\sigma = 10$ min, $n = 50$, $z\ (95\%) = 1.96$

**2** **Compute the standard error**

$\sigma/\sqrt{n} = 10 / \sqrt{50} \approx 1.41$

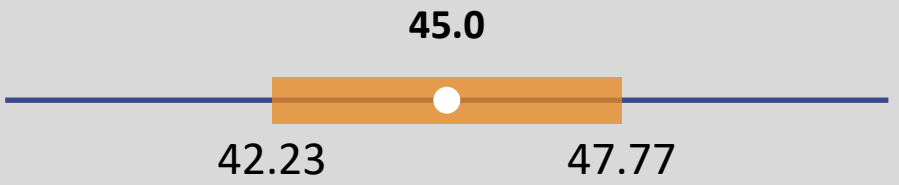
**3** **Compute the margin of error**

$z \times SE = 1.96 \times 1.41 \approx 2.77$

**4** **Build the interval**

$CI = 45 \pm 2.77 = (42.23, 47.77)$

**Visualizing the Interval**




*95% confident the true average wait time lies within the shaded band.*

***Interpretation:*** *We are 95% confident that the true average wait time for all customers falls between 42.23 and 47.77 minutes.*

# Tests of Hypothesis

A statistical hypothesis is an assumption about a population parameter. Hypothesis testing (or test of significance) is the decision-making procedure that uses sample data to decide whether to accept or reject that assumption.

**1** **Drug testing**

Is a new drug really effective in curing a disease?

**2** **Quality control**

Is a manufacturing process working properly?

**3** **Fairness checks**

Is a given coin (or die) biased?

# Null Hypothesis vs. Alternative Hypothesis

## Null Hypothesis $H_0$

$$H_0 : \mu = \mu_0$$

A definite statement about the population parameter — Such a hypothesis is usually a hypothesis of no-difference.

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*$\mu_0$ = the value assumed/claimed for the population characteristics.*

## Alternative Hypothesis $H_1$

*Contradicts $H_0$ — rejecting $H_0$ implies accepting $H_1$*

$$H_1 : \mu \neq \mu_0$$

$$H_1 : \mu > \mu_0$$

$$H_1 : \mu < \mu_0$$

# Procedure for Testing a Hypothesis

## 1

### Null Hypothesis

Set up $H_o$ based on the nature of the problem and data involved.

## 2

### Alternative Hypothesis

Set up $H_1$, deciding one-tailed or two-tailed test.

## 3

### Level of Significance

Select $\alpha$ (usually 5% or 1%) in advance, based on reliability needed.

# Level of Significance (α)

The level of significance α is the maximum probability of rejecting a *true* null hypothesis $H_0$ — i.e., the risk of error we are willing to accept. It is fixed **before** the test is carried out, so results don't influence the choice.

## 5%

*( 0.05 )*

≈ 5 in 100 cases we'd wrongly reject a true $H_0$ — 95% confident the decision is right.

## 1%

*( 0.01 )*

Only 1 in 100 cases of wrongly rejecting $H_0$ — 99% confident the decision is right.

## Also used

*( 2%, 0.5% )*

Other levels appear occasionally, though 5% and 1% dominate practice.

*Level of significance is also known as the size of the test.*

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Select α (usually 5% or 1%) in advance, based on reliability needed.

## 4

### Test Statistic

Which one to use **Z or t test**— depends on whether the population S.D. is known and how large the sample is.

# Step 4 in Focus: Choosing the Test Statistic

Step 4 asks us to compute a test statistic. Which one to use — **Z** or **t** — depends on whether the population S.D. is known and how large the sample is.

| Sample size | **Population S.D. (σ) is known** | **Population S.D. (σ) is unknown** |
|---|---|---|
| **N ≥ 30 (Any Population)** | **z- test** | **z - test** |
| **N < 30 (Any Population)** | **z- test** | **t-test** |

# Procedure for Testing a Hypothesis

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| **Null Hypothesis** | **Alternative Hypothesis** | **Level of Significance** | **Test Statistic** | **Conclusion** |
| Set up $H_o$ based on the nature of the problem and data involved. | Set up $H_1$, deciding one-tailed or two-tailed test. | Select α (usually 5% or 1%) in advance, based on reliability needed. | Which one to use **Z or t test**— depends on whether the population S.D. is known and how large the sample is. | Critical value decision is taken for accepting or rejecting $H_0$ .If the computed value < critical value, we accept $H_0$ , otherwise we reject $H_0$. |

# Errors of Sampling

In practice we decide to accept or reject a hypothesis after examining only a sample — two kinds of wrong decisions are possible:

## TYPE I ERROR

**Reject $H_0$ when it is true.**

Probability of Type I error = α (the level of significance).

Also called the producer's risk.

## TYPE II ERROR

**Accept $H_0$ when it is false ($H_1$ is true).**

Probability of Type II error = β.

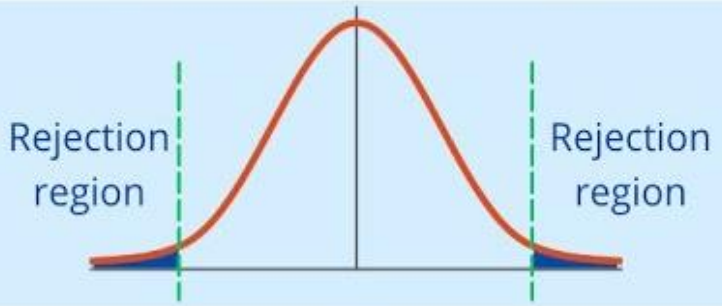
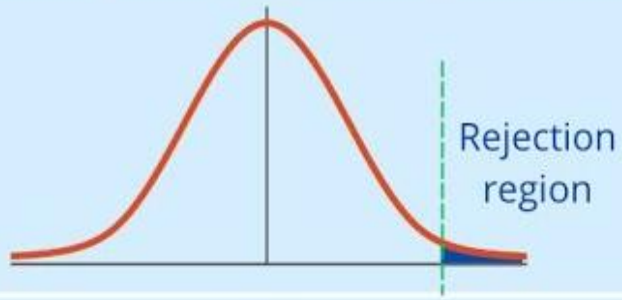
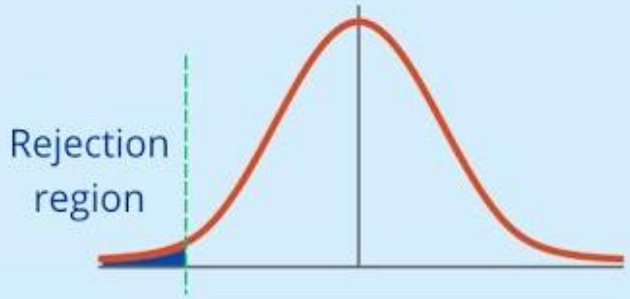
Also called the consumer's risk.

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true}) = P(\text{Rejecting a good lot})$$

$$\beta = P(\text{Accept } H_0 \mid H_0 \text{ false}) = P(\text{Accepting a bad lot})$$

**Key point:** The only way to reduce both types of error simultaneously is to increase the sample size,

# One-Tailed vs. Two-Tailed Tests

| Test Type | Hypotheses | Rejection Region |
|---|---|---|
| Two-Tailed | $H_0 : \mu = \mu_0$<br>$H_1 : \mu \neq \mu_0$ |  |
| Right-Tailed | $H_0 : \mu \leqslant \mu_0$<br>$H_1 : \mu > \mu_0$ |  |
| Left-Tailed | $H_0 : \mu \geqslant \mu_0$<br>$H_1 : \mu < \mu_0$ |  |

# Z-Test: Sample size is n>30

| 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|
| **Null Hypothesis** | **Alternative Hypothesis** | **Level of Significance** | **Test Statistic** | **Conclusion** |
| Set up $H_o$ based on the nature of the problem and data involved. | Set up $H_1$, deciding one-tailed or two-tailed test. | Select α (usually 5% or 1%) in advance, based on reliability needed. | Compute $Z = (t - E(t)) / S.E.(t)$ under $H_o$.<br>t=Sample Statistic<br>E(t)=Expected Statistic | Compare $\|Z\|$ with critical value Zα: accept $H_o$ if $\|Z\| < Z\alpha$, else reject. |

# Critical Values of Z

| Level of Significance α | 1% (0.01) | 5% (0.05) | 10% (0.1) |
|---|---|---|---|
| **Two-Tailed Test \|Zα\|** | 2.58 | 1.96 | 1.645 |
| **Right-Tailed Test Zα** | 2.33 | 1.645 | 1.28 |
| **Left-Tailed Test Zα** | −2.33 | −1.645 | −1.28 |

## Decision Rule

- If $|Z| <$ critical value $Z\alpha \rightarrow$ accept $H_o$ (difference not significant).
- If $|Z| >$ critical value $Z\alpha \rightarrow$ reject $H_o$ (difference significant) at that level.
- Note: $H_o$ can be rejected outright, without naming a level, if $|Z| > 3$.

# Z-Test: Sample size is n>30

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# Four Tests for Large Samples

Under large-sample tests, we study four important tests of significance:

1

### Single Proportion

Tests whether a sample proportion differs significantly from an assumed population proportion p.

2

### Difference of Proportions

Tests whether two sample proportions differ significantly from each other.

3

### Single Mean

Tests whether a sample mean differs significantly from a specified population mean $\mu_o$.

4

### Difference of Means

Tests whether the means of two independent samples differ significantly from each other.

**All four use the same Z framework:** `Z = (Observed − Expected) / S.E.,` compared against the critical value Zα for the chosen level of significance.

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**All four use the same Z framework:** `Z = (Observed − Expected) / S.E.,` compared against the critical value $Z_\alpha$ for the chosen level of significance.

# Z-Test of Significance for a Single Mean

For a random sample of size n (n > 30) with sample mean $\bar{x}$, drawn from a population of mean μ:

$$H_0 : \bar{x} = \mu \qquad H_1 : \bar{x} \neq \mu \quad (\text{or} \quad \bar{x} > \mu \quad / \quad \bar{x} < \mu)$$

### Case I — population S.D. (σ) known

$$S.E.(\bar{x}) = \sigma / \sqrt{n}$$

$$Z = (\bar{x} - \mu) / (\sigma / \sqrt{n})$$

*Use the population standard deviation directly.*

### Case II — population S.D. unknown

$$S.E.(\bar{x}) = s / \sqrt{n}$$

$$Z = (\bar{x} - \mu) / (s / \sqrt{n})$$

*Use the sample standard deviation s in its place.*

| Level of Significance α | 1% (0.01) | 5% (0.05) | 10% (0.1) |
|---|---|---|---|
| **Two-Tailed Test \|Zα\|** | 2.58 | 1.96 | 1.645 |
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# Example

An oceanographer wants to check whether the depth of the ocean in a region is 57.4 fathoms, as previously recorded. Readings at 40 random locations gave a mean of 59.1 fathoms with S.D. 5.2 fathoms. Test at $\alpha = 0.05$.

**Given** $\bar{x} = 59.1 \quad \sigma = 5.2 \quad n = 40 \quad \alpha = 0.05$

**1. Null Hypothesis** $H_0 : \mu = 57.4$

**2. Alternative Hypothesis** $H_1 : \mu \neq 57.4$ (two-tailed)

**3. Level of Significance** $\alpha = 0.05$

**4. Test Statistic**

$$\text{Calculated } Z = (\bar{x} - \mu)/(\sigma/\sqrt{n})$$
$$= (59.1 - 57.4)/(5.2/\sqrt{40}) = 2.067$$

$$\text{Tabulated } Z_{0.05} = 1.96$$

**Conclusion:** Calculated Z = 2.067 > Tabulated $Z_{0.05} = 1.96$

**→ $H_0$ is rejected. The depth is significantly different from 57.4 fathoms.**

# What Is a T-Test?

- A t-test is a statistical method used to determine if there is a **significant difference between the means of two groups** when the population standard deviation is unknown.
- In t-test statistics, the sample data is a subset of the two groups used to draw conclusions about the groups as a whole.

- ***Example:*** *A teacher wants to know whether the average score of a class is different from the school average. A t-test can be used to make this comparison.*

**THREE COMMON TYPES**

1. **One-Sample T-Test**
   Sample vs. a known population mean

2. **Independent Samples**
   Two separate, unrelated groups

3. **Paired Samples**
   Same subjects, measured twice

# 1 One-Sample T-Test

Used when we have one data set for a sample and need to determine whether it belongs to a particular population. The population mean must be known.




***Example:*** *Determining whether the average weight of a sample of mangoes differs from the known average weight of all mangoes on a farm.*

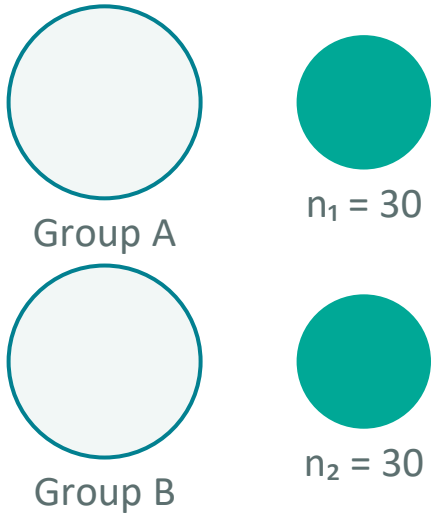
**FORMULA**

$$t = (\bar{x} - \mu) / (s / \sqrt{n})$$

- $\bar{x}$ = sample mean
- $\mu$ = population mean
- $s$ = sample standard deviation
- $n$ = sample size

# 2 Independent Samples T-Test

- Used to compare the statistical means of two independent, unrelated samples or groups.
- Helps determine whether there is a significant difference between the means of the two groups.
- If there is a significant difference, the groups likely have different population means; otherwise, they share the same population mean.




**FORMULA**

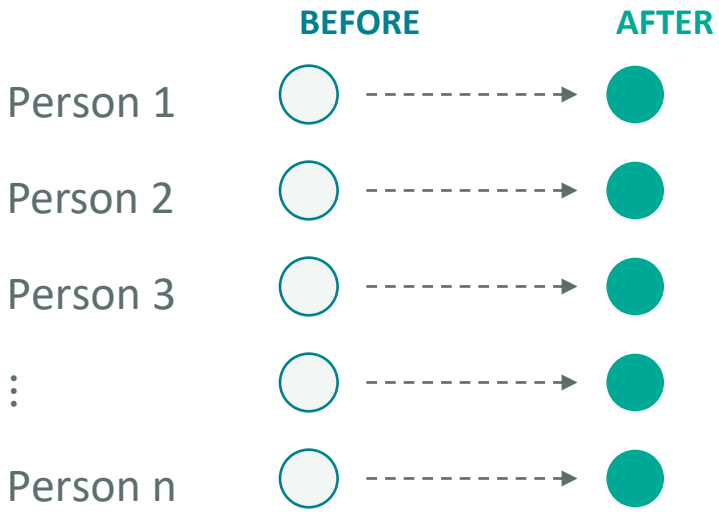
$$t = \frac{(\bar{x} - \bar{y})}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}}$$

- $\bar{x}$ , $\bar{y}$ = sample means
- $s^2 = \frac{1}{n_1+n_2-2}[\sum(x_i - \bar{x})^2 + \sum(y_i - \bar{y})^2]$
- $n_1$, $n_2$ = sample sizes

***Example:*** *Comparing mean scores of athletes who followed a training camp (Team A) vs. those who did not (Team B).*

# 3 Paired Samples T-Test

Used when observations are related or collected from the same subjects at different times — for example, before and after a training program.




***Example:*** *Comparing the average scores of athletics team players before and after a training program.*

**FORMULA**

$$t = (\bar{d} - \mu^d) / (s^d / \sqrt{n})$$

- $\bar{d}$ = mean of the differences
- $\mu^d$ = population mean difference (usually 0)
- $s^d$ = standard deviation of the differences
- n = number of pairs

# How to Perform a T-Test

*The general six-step procedure*

### 1 Define hypotheses

State the null ($H_0$) and alternative ($H_1$) hypotheses.

### 2 Select significance level

Choose $\alpha$, usually 0.05.

### 3 Apply the t-test formula

Use the formula matching your test type.

### 4 Determine degrees of freedom

One-sample & paired: $df = n - 1$. Independent: $df = n_1 + n_2 - 2$.

### 5 Find the p-value

Use a t-distribution table.

### 6 Draw a conclusion

If p-value $< \alpha$, reject $H_0$. If p-value $\geq \alpha$, fail to reject $H_0$.

# Degrees of Freedom (df)

The degrees of freedom tell the t-test how much “information” is in your data. The fewer people you measure, the more likely random fluctuation creates extreme results — df adjusts the t-distribution to account for this.

**One-Sample T-Test**

$$df = n - 1$$

**Independent Samples T-Test**

$$df = n_1 + n_2 - 2$$

**Paired Samples T-Test**

$$df = n - 1$$

**Note:** For a one-sample test, n is the number of observations; for a paired test, it is the number of pairs. The equal-variance independent t-test uses both group sizes — Welch’s t-test uses a different df formula.

# Example: One-Sample T-Test

A random sample of 10 boys had the following I.Q.'s: $\mathbf{70, 120, 110, 101, 88, 83, 95, 98, 107, 100}$

**(a)** Do these data support the assumption of a population mean I.Q. of 100?

**(b)** Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

*Approach: use a one-sample t-test since the population standard deviation is unknown — it must be estimated from the sample.*

# Example: One-Sample T-Test

## Compute the Mean and Standard Deviation

*Since the sample mean and S.D. aren't given directly, they must be calculated first.*

| $x$ | $x - \bar{x}$ | $(x - \bar{x})^2$ |
|---|---|---|
| 70 | −27.2 | 739.84 |
| 120 | 22.8 | 519.84 |
| 110 | 12.8 | 163.84 |
| 101 | 3.8 | 14.44 |
| 88 | −9.2 | 84.64 |
| 83 | −14.2 | 201.64 |
| 95 | −2.2 | 4.84 |
| 98 | 0.8 | 0.64 |
| 107 | 9.8 | 96.04 |
| 100 | 2.8 | 7.84 |
| **Σ = 972** | | **1833.60** |

**MEAN**

$$\bar{x} = \Sigma x / n = 972 / 10 = 97.2$$

**We know that sample variance**

$$s^2 = \Sigma(x_i - \bar{x})^2 / (n-1) = 1833.60 / 9 = 203.73$$

**Sample STANDARD DEVIATION**

$$s = \sqrt{203.73} = 14.27$$

# Example: One-Sample T-Test

**(a)** Do these data support the assumption of a population mean I.Q. of 100?

$\bar{x} = 97.2$ $s = 14.27$ $n = 10$ $\alpha = 0.05$

**1. Null Hypothesis** $H_0 : \mu = 100$

**2. Alternative Hypothesis** $H_1 : \mu \neq 100$ (two-tailed)

**3. Level of Significance** $\alpha = 0.05$

**4. Test Statistic** Calculated $t = (\bar{x} - \mu) / (s / \sqrt{n})$

$= (97.2 - 100) / (14.27/\sqrt{10}) = |-0.62| = 0.62$

Tabulated t for d.f. at $\alpha$ (0.05) (**where d. f= n-1 = 10-1 =9**)

**= 2.262**

**Conclusion:** Calculated $|t| = 0.62$ < Tabulated $t_{(9,0.05)} = 2.262$

→ **$H_0$ is accepted. The data support the assumption of a population mean I.Q. of 100.**

# Example: One-Sample T-Test

**(b)** Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

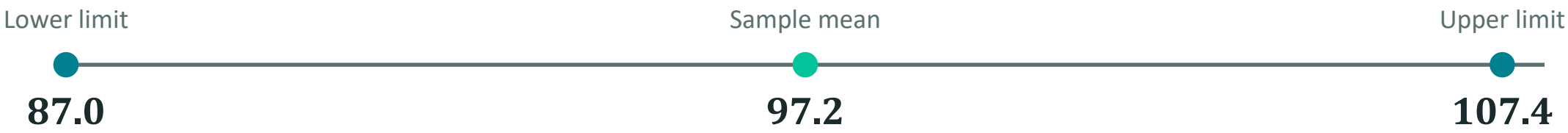
$\bar{x} = 97.2 \quad s = 14.27 \quad n = 10 \quad \alpha = 0.05$

$\alpha = 0.05$ (5%) - The 95% Confidence Interval

A reasonable range for the mean I.Q. of samples of 10 boys, using the 95% confidence limits:

$$\bar{x} \pm t_{0.05} \cdot s/\sqrt{n} = 97.2 \pm 2.26 \times (14.27/\sqrt{10}) = 97.2 \pm 2.26 \times 4.512 = 97.2 \pm 10.198$$

$$= 107.4 \text{ and } 87.0$$




**Conclusion:** The 95% confidence limits within which the mean I.Q. values of samples of 10 boys will lie is (87, 107.40).

# Example: Independent Samples T-Test

Gain in weight (lbs) of pigs fed on two diets, A and B. Test if the diets differ significantly in their effect on weight increase.

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Diet A** | 25 | 32 | 30 | 34 | 24 | 14 | 32 | 24 | 30 | 31 | 35 | 25 | – | – | – |
| **Diet B** | 44 | 34 | 22 | 10 | 47 | 31 | 40 | 30 | 32 | 35 | 18 | 21 | 35 | 29 | 22 |

# Example: Independent Samples T-Test

$n_1 = 12,\ \Sigma x = 336$

$\bar{x} = 336/12 = 28$

$n_2 = 15,\ \Sigma y = 450$

$\bar{y} = 450/15 = 30$

| x | $x - \bar{x}$ | $(x - \bar{x})^2$ | y | $(y - \bar{y})$ | $(y - \bar{y})^2$ |
|---|---|---|---|---|---|
| 25 | −3 | 9 | 44 | 14 | 196 |
| 32 | 4 | 16 | 34 | 4 | 16 |
| 30 | 2 | 4 | 22 | −8 | 64 |
| 34 | 6 | 36 | 10 | −20 | 400 |
| 24 | −4 | 16 | 47 | 17 | 289 |
| 14 | −14 | 196 | 31 | 1 | 1 |
| 32 | 4 | 16 | 40 | 10 | 100 |
| 24 | −4 | 16 | 30 | 0 | 0 |
| 30 | 2 | 4 | 32 | 2 | 4 |
| 31 | 3 | 9 | 35 | 5 | 25 |
| 35 | 7 | 49 | 18 | −12 | 144 |
| 25 | −3 | 9 | 21 | −9 | 81 |
| | | | 35 | 5 | 25 |
| | | | 29 | −1 | 1 |
| | | | 22 | −8 | 64 |
| **336** | | **380** | **450** | | **1410** |

# Example: Independent Samples T-Test

**1. Null Hypothesis** $H_0 : \mu_1 = \mu_2$ — no significant difference between the mean weight gain from diets A and B.

**2. Alternative Hypothesis** $H_1 : \mu \neq \mu_2$ (two-tailed)

**3. Level of Significance** $\alpha = 0.05$

**4. Test Statistic**

$$\text{Calculated } t = \frac{(\bar{x} - \bar{y})}{\sqrt{\frac{s^2}{n_1} + \frac{s^2}{n_2}}} \quad \text{where } s^2 = \frac{1}{n_1 + n_2 - 2}\left[\sum(x_i - \bar{x})^2 + \sum(y_i - \bar{y})^2\right]$$

$\mathbf{S^2}$ = [380 + 1410] / (12+15−2) =71.6

Calculated $\mathbf{|t| = (30-28) / \sqrt{(71.6/12 + 71.6/15)} = 2/\sqrt{10.74} = 0.609}$

Tabulated t for d.f. at α (0.05) (**where d. f= $n_1 + n_2 - 2 = 12+15-2 = 25$)**

Tabulated t for **25** at α (0.05) **= 2.06**

**Conclusion:** Calculated t = 0.609 < Tabulated $t_{(25,0.05)}$ = 2.06 → **$H_0$ is accepted. The two diets do not differ significantly in their effect on weight increase.**

# Example: Paired Samples T-Test

Scores obtained in a shooting competition by 10 soldiers before and after intensive training are given below:

| Soldier | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Before | 67 | 24 | 57 | 55 | 63 | 54 | 56 | 68 | 33 | 43 |
| After | 70 | 38 | 58 | 58 | 56 | 67 | 68 | 75 | 42 | 38 |

**1. Null Hypothesis** $H_0 : \mu_1 = \mu_2$, i.e. $\mu = 0$. Training has no significant effect.

**2. Alternative Hypothesis** $H_1 : \mu \neq \mu_2$ (two-tailed)

**3. Level of Significance** $\alpha = 0.05$

**4. degrees of freedom** $n - 1 = 9$ d.f.

**5. The t-test statistic** $t = (\bar{d} - \mu^d) / (s^d / \sqrt{n})$

# Example: Paired Samples T-Test

**5. The t-test statistic** $\mathbf{t = (\bar{d} - \mu^d) / (s^d / \sqrt{n})}$

**a. Differences** $d_i$=before - after

**b.** $\bar{d}$ $$\frac{1}{n}\sum_{i=1}^{n} d_i = \frac{1}{10}\sum_{i=1}^{10} d_i = \frac{-50}{10} = -5$$

**c.** $s^2$ $$\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2 = \frac{482}{9} = 53.555$$

**d.** $s$ $$\sqrt{53.5555} = 7.32$$

$\mathbf{= (\bar{d} - \mu^d) / (s^d / \sqrt{n})}$

**e.** calculated t $= (-5 - 0)/(7.32/\sqrt{10})$

$= -2.16$

**e.** tabulated t $t_{(9,0.05)} = 1.83$

| Soldier | Before | After | d = B − A | $d_i - \bar{d}$ | $(d_i - \bar{d})^2$ |
|---|---|---|---|---|---|
| 1 | 67 | 70 | −3 | **2** | **4** |
| 2 | 24 | 38 | −14 | **-9** | **81** |
| 3 | 57 | 58 | −1 | **4** | **16** |
| 4 | 55 | 58 | −3 | **2** | **4** |
| 5 | 63 | 56 | 7 | **12** | **144** |
| 6 | 54 | 67 | −13 | **-8** | **64** |
| 7 | 56 | 68 | −12 | **-7** | **49** |
| 8 | 68 | 75 | −7 | **-2** | **4** |
| 9 | 33 | 42 | −9 | **-4** | **16** |
| 10 | 43 | 38 | 5 | **10** | **100** |

**Conclusion:** Calculated |t| = 2.16 > Tabulated $t_{(9,0.05)}$ = 1.83 → **$H_0$ is rejected. Intensive training is useful.**

# ANOVA: Analysis of Variance

- Tests whether there is a difference between two or more population means
- For just two samples, a z-test (large samples) or t-test (small samples) is used instead
- Running many t-tests to compare several groups inflates the Type I error rate
- ANOVA compares all group means at once, avoiding that problem

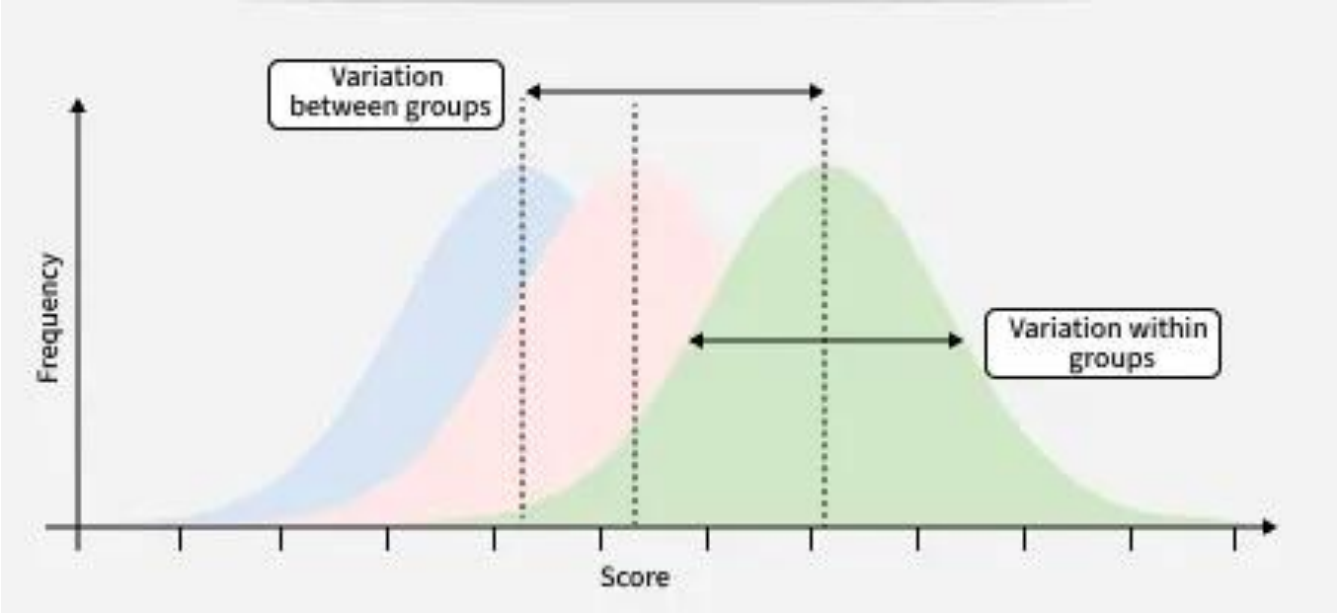
# What is ANOVA & Basic Principle

- A method for testing differences among groups for homogeneity
- Make two estimates of population variance

  1. one based on between-samples variance  2. the other based on within-samples variance.

- These two estimates are then compared using the **F-test**, where

$$F = \frac{\textit{Estimate of population variance based on between} - \textit{samples variance}}{\textit{Estimate of population variance based on within} - \textit{samples variance}}$$



# One-way (or single factor) ANOVA

- Under the one-way ANOVA, we consider only one factor and then observe that the reason for said factor to be important is that several possible types of samples can occur within that factor. We then determine if there are differences within that factor.
- The technique involves the following steps:

1. Obtain the mean of each sample i.e., obtain $\bar{X}_1, \bar{X}_2 \ldots \bar{X}_k$ when there are *k* samples.
2. Work out the mean of the sample means as follows. $\bar{\bar{X}} = \frac{\bar{X}_1 + \bar{X}_2 + \cdots + \bar{X}_k}{\text{No. of samples } (k)}$
3. Calculate the sum of squares for variance between the samples (or SS between). Symbolically, this can be written. $\textbf{SS between} = n_1 (\bar{X}_1 - \bar{\bar{X}})^2 + n_2 (\bar{X}_2 - \bar{\bar{X}})^2 + \cdots + n_k (\bar{X}_k - \bar{\bar{X}})^2$
   where $n_1, n_2, \ldots n_k$ number of items in the corresponding sample
4. Divide the result of the (3) step by the degrees of freedom between the samples to obtain variance or mean square (MS) between samples. Symbolically, this can be written $\textbf{MS between} = \frac{\textbf{SS between}}{(k-1)}$ where $k - 1 = \text{d.f.}$
5. Calculate the sum of squares for variance between the samples (or SS between). Symbolically, this can be written. $\textbf{SS within} = \sum(X_{1i} - \bar{X}_1)^2 + \sum(X_{2i} - \bar{X}_1)^2 + \ldots + \sum(X_{ki} - \bar{X}_1)^2$ where $i = 1, 2, 3 \ldots$

# One-way (or single factor) ANOVA

6. Divide the result of (5) step by the degrees of freedom within samples to obtain the variance or mean square (MS) within samples. Symbolically, this can be written. $MS \text{ within} = \frac{\text{SS within}}{(n-k)}$

where $(n - k)$ represents degrees of freedom within samples,
$n$ = total number of items in all the samples i.e., $n_1 + n_2 + \ldots + n_k$ *where* $k$ = number of samples.

7. Check, the sum of squares of deviations for total variance. Symbolically, this can be written

$$\boldsymbol{SS\ for\ total\ variance} = \sum\left(X_{ij} - \bar{\bar{X}}\right)^2 \text{ where } i = 1, 2, 3, \ldots \ \& \ \ j = 1, 2, 3, \ldots$$

This total should be equal to the total of the result of the (3) and (5) steps explained above i.e.,

$$SS\ for\ total\ variance = SS\ between + SS\ within.$$

The degrees of freedom for total variance will be equal to the number of items in all samples minus one i.e., (n – 1). The degrees of freedom for between and within must add up to the degrees of freedom for total variance i.e., **(n – 1) = (k – 1) + (n – k)**

8. Finally, F-ratio may be worked out as under: $\boldsymbol{F - ratio} = \frac{\boldsymbol{MS\ between}}{\boldsymbol{MS\ within}}$

The F-ratio is used to determine whether differences among sample means are significant or due to chance.
Calculated F < Table F: Difference is insignificant → accept the null hypothesis.
Calculated F ≥ Table F: Difference is significant → reject the null hypothesis.
Higher F value: Stronger evidence that the differences are significant

# Setting Up Analysis Of Variance Table

For the sake of convenience the information obtained through various steps stated above can be put as under:

| Source of variation | Sum of squares (SS) | Degrees of freedom (d.f.) | Mean Square (MS) | F-ratio |
|---|---|---|---|---|
| Between samples or categories | $\mathbf{SS\ between} = n_1\left(\bar{X}_1 - \bar{\bar{X}}\right)^2 + \cdots + n_k\left(\bar{X}_k - \bar{\bar{X}}\right)^2$ | $k-1$ | $MS_{\text{between}} = \frac{SS_{\text{between}}}{k-1}$ | $\frac{MS_{\text{between}}}{MS_{\text{within}}}$ |
| Within samples or categories | $\mathbf{SS\ within} = \sum(X_{1i} - \bar{X}_1)^2 + \ldots + \sum(X_{ki} - \bar{X}_1)^2$ | $n-k$ | $MS_{\text{within}} = \frac{SS_{\text{within}}}{n-k}$ | — |
| Total | | $n-1$ | — | — |

# Example

Set up an analysis of variance table for the following per acre production data for three varieties of wheat, each grown on 4 plots and state if the variety differences are significant.

| Plot of land | Per acre production data: Variety of wheat A | Per acre production data: Variety of wheat B | Per acre production data: Variety of wheat C |
|---|---|---|---|
| **1** | 6 | 5 | 5 |
| **2** | 7 | 5 | 4 |
| **3** | 3 | 3 | 3 |
| **4** | 8 | 7 | 4 |

1. Obtain the mean of each sample i.e., obtain $\bar{X}_1, \bar{X}_2 \ldots \bar{X}_k$ when there are $k$ samples.

$$\bar{X}_1 = \frac{6+7+3+8}{4} = 6$$

$$\bar{X}_2 = \frac{5+5+3+7}{4} = 5$$

$$\bar{X}_3 = \frac{5+4+3+4}{4} = 4$$

1. Obtain the mean of each sample i.e., obtain $\bar{X}_1, \bar{X}_2 \ldots \bar{X}_k$ when there are $k$ samples.

$$\bar{X}_1 = \frac{6 + 7 + 3 + 8}{4} = 6$$

$$\bar{X}_2 = \frac{5 + 5 + 3 + 7}{4} = 5$$

$$\bar{X}_3 = \frac{5 + 4 + 3 + 4}{4} = 4$$

2. Work out the mean of the sample means as follows. $\bar{\bar{X}} = \dfrac{\bar{X}_1 + \bar{X}_2 + \cdots + \bar{X}_k}{\text{No. of samples } (k)}$

$$\bar{\bar{X}} = \frac{\bar{X}_1 + \bar{X}_2 + \bar{X}_3}{k}$$

$$= \frac{6 + 5 + 4}{3} = 5$$

3. Calculate the sum of squares for variance between the samples (or SS between). Symbolically, this can be written. $\mathbf{SS\ between} = \boldsymbol{n_1}\left(\bar{\boldsymbol{X}}_{\boldsymbol{1}} - \bar{\bar{\boldsymbol{X}}}\right)^{\boldsymbol{2}} + \boldsymbol{n_2}\left(\bar{\boldsymbol{X}}_{\boldsymbol{2}} - \bar{\bar{\boldsymbol{X}}}\right)^{\boldsymbol{2}} + \cdots + \boldsymbol{n_k}\left(\bar{\boldsymbol{X}}_{\boldsymbol{k}} - \bar{\bar{\boldsymbol{X}}}\right)^{\boldsymbol{2}}$

$$SS \text{ between} = n_1\left(\bar{X}_1 - \bar{\bar{X}}\right)^2 + n_2\left(\bar{X}_2 - \bar{\bar{X}}\right)^2 + n_3\left(\bar{X}_3 - \bar{\bar{X}}\right)^2$$
$$= 4(6-5)^2 + 4(5-5)^2 + 4(4-5)^2$$
$$= 4 + 0 + 4$$
$$= 8$$

4. Divide the result of the (3) step by the degrees of freedom between the samples to obtain variance or mean square (MS) between samples. Symbolically, this can be written $\boldsymbol{MS\ between} = \frac{\boldsymbol{SS\ between}}{(\boldsymbol{k-1})}$

$$MS\ between = \frac{8}{(3-1)} = \frac{8}{2} = 4$$

5. Calculate the sum of squares for variance between the samples (or SS between). Symbolically, this can be written. **SS within** $= \sum(X_{1i} - \bar{X}_1)^2 + \sum(X_{2i} - \bar{X}_1)^2 + \ldots + \sum(X_{ki} - \bar{X}_1)^2$ where $i = 1, 2, 3 \ldots$

$$SS \text{ within} = \sum\left(X_{1i} - \bar{X}_1\right)^2 + \sum\left(X_{2i} - \bar{X}_2\right)^2 + \sum\left(X_{3i} - \bar{X}_3\right)^2, \qquad i = 1, 2, 3, 4$$

$$= \{(6-6)^2 + (7-6)^2 + (3-6)^2 + (8-6)^2\}$$
$$+ \{(5-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2\}$$
$$+ \{(5-4)^2 + (4-4)^2 + (3-4)^2 + (4-4)^2\}$$
$$= \{0 + 1 + 9 + 4\} + \{0 + 0 + 4 + 4\} + \{1 + 0 + 1 + 0\}$$
$$= 14 + 8 + 2$$
$$= 24$$

6. Divide the result of (v) step by the degrees of freedom within samples to obtain the variance or mean square (MS) within samples. Symbolically, this can be written. $MS \text{ within} = \frac{\text{SS within}}{(n-k)}$

$$MS \text{ within} = \frac{24}{(12-3)} = \frac{24}{9} = 2.67$$

7. Check, the sum of squares of deviations for total variance. Symbolically, this can be written

$$\boldsymbol{SS\ for\ total\ variance} = \sum\left(\boldsymbol{X_{ij}} - \boldsymbol{\bar{\bar{X}}}\right)^{\boldsymbol{2}} \quad \text{where } i = 1, 2, 3, \ldots \ \&\ \ j = 1, 2, 3, \ldots$$

This total should be equal to the total of the result of the (3) and (5) steps explained above i.e.,

$$SS\ for\ total\ variance = SS\ between + SS\ within.$$

$$SS \text{ for total variance} = \sum\left(X_{ij} - \bar{\bar{X}}\right)^2 \qquad i = 1, 2, 3\ldots$$

$$j = 1, 2, 3\ldots$$

$$= (6-5)^2 + (7-5)^2 + (3-5)^2 + (8-5)^2 + (5-5)^2 + (5-5)^2 + (3-5)^2 + (7-5)^2 + (5-5)^2 + (4-5)^2 + (3-5)^2 + (4-5)^2$$

$$= 1 + 4 + 4 + 9 + 0 + 0 + 4 + 4 + 0 + 1 + 4 + 1$$

$$= 32$$

Alternatively, it (*SS* for total variance) can also be worked out thus:

*SS* for total = *SS* between + *SS* within

$$= 8 + 24$$

$$= 32$$

8. Finally, F-ratio may be worked out as under: $\boldsymbol{F - ratio} = \dfrac{\boldsymbol{MS\ between}}{\boldsymbol{MS\ within}}$

The F-ratio is used to determine whether differences among sample means are significant or due to chance.

Calculated F < Table F: Difference is insignificant → accept the null hypothesis.

Calculated F ≥ Table F: Difference is significant → reject the null hypothesis.

Higher F value: Stronger evidence that the differences are significant

8. Finally, F-ratio may be worked out as under: $\boldsymbol{F - ratio = \frac{MS\ between}{MS\ within}}$

The F-ratio is used to determine whether differences among sample means are significant or due to chance.
Calculated F < Table F: Difference is insignificant → accept the null hypothesis.
Calculated F ≥ Table F: Difference is significant → reject the null hypothesis.
Higher F value: Stronger evidence that the differences are significant

| **Source of variation** | **SS** | **d.f.** | **MS** | **F-ratio** | **5% F-limit (from the F-table)** |
|---|---|---|---|---|---|
| **Between sample** | 8 | (3 - 1) = 2 | 8 / 2 = 4.00 | 4.00 / 2.67 = 1.5 | F(2, 9) = 4.26 |
| **Within sample** | 24 | (12 - 3) = 9 | 24 / 9 = 2.67 | | |
| **Total** | 32 | (12 - 1) = 11 | | | |

- The above table shows that the calculated value of *F* is 1.5 which is less than the table value of 4.26 at 5% level with d.f. being $v_1 = 2$(column) and $v_2 = 9$(row) and hence could have arisen due to chance.
- This analysis supports the null-hypothesis of no difference is sample means.

# Two-way ANOVA

- A **two-way ANOVA** (Analysis of Variance) is a statistical test used to determine how two categorical independent variables (factors) simultaneously affect a single continuous dependent variable.

- **Two-way ANOVA** is used when data are affected by **two factors**. For example, agricultural output may depend on:
  **Factor A:** Type of seed
  **Factor B:** Type of fertilizer

- We want to determine whether the variation in output is due to **Seed**, **Fertilizer**, or **random/error variation**.

# Two-way ANOVA:

| Source of variation | Sum of squares (SS) | Degrees of freedom (d.f.) | Mean square (MS) | F-ratio |
|---|---|---|---|---|
| **Between columns treatment** | $\sum \frac{(T_j)^2}{n_j} - \frac{(T)^2}{n}$ | $(c-1)$ | $\frac{SS \text{ between columns}}{(c-1)}$ | $\frac{MS \text{ between columns}}{MS \text{ residual}}$ |
| **Between rows treatment** | $\sum \frac{(T_i)^2}{n_i} - \frac{(T)^2}{n}$ | $(r-1)$ | $\frac{SS \text{ between rows}}{(r-1)}$ | $\frac{MS \text{ between rows}}{MS \text{ residual}}$ |
| **Residual or error** | Total $SS$ $-$ ($SS$ between columns $+$ $SS$ between rows) | $(c-1)(r-1)$ | $\frac{SS \text{ residual}}{(c-1)(r-1)}$ | |
| **Total** | $\sum X_{ij}^2 - \frac{(T)^2}{n}$ | $(c \cdot r - 1)$ | | |

*T = total of the values of individual items*
*c* = number of columns
*r* = number of rows
*SS* residual = Total *SS* – (*SS* between columns + *SS* between rows).

# Two-way ANOVA:

Set up an analysis of variance table for the following two-way design results:

**Per Acre Production Data of Wheat**

(*in metric tonnes*)

| *Varieties of seeds* | *A* | *B* | *C* |
|---|---|---|---|
| Varieties of fertilizers | | | |
| *W* | 6 | 5 | 5 |
| *X* | 7 | 5 | 4 |
| *Y* | 3 | 3 | 3 |
| *Z* | 8 | 7 | 4 |

Also state whether variety differences are significant at 5% level.

# Two-way ANOVA:

Step (i) $T = 60, n = 12, \therefore$ Correction factor $= \dfrac{(T)^2}{n} = \dfrac{60 \times 60}{12} = 300$

Step (ii) Total $SS = (36 + 25 + 25 + 49 + 25 + 16 + 9 + 9 + 9 + 64 + 49 + 16) - \left(\dfrac{60 \times 60}{12}\right)$

$= 332 - 300$

$= 32$

Step (iii) $SS$ between columns treatment $= \left[\dfrac{24 \times 24}{4} + \dfrac{20 \times 20}{4} + \dfrac{16 \times 16}{4}\right] - \left[\dfrac{60 \times 60}{12}\right]$

$= 144 + 100 + 64 - 300$

$= 8$

Step (iv) $SS$ between rows treatment $= \left[\dfrac{16 \times 16}{3} + \dfrac{16 \times 16}{3} + \dfrac{9 \times 9}{3} + \dfrac{19 \times 19}{3}\right] - \left[\dfrac{60 \times 60}{12}\right]$

$= 85.33 + 85.33 + 27.00 + 120.33 - 300$

$= 18$

Step (v) $SS$ residual or error = Total $SS$ – ($SS$ between columns + $SS$ between rows)

$= 32 - (8 + 18)$

$= 6$

# Two-way ANOVA:

| *Source of variation* | *SS* | *d.f.* | *MS* | *F-ratio* | *5% F-limit (or the tables values)* |
|---|---|---|---|---|---|
| Between columns (i.e., between varieties of seeds) | 8 | $(3-1)=2$ | $8/2=4$ | $4/1=4$ | $F(2,6)=5.14$ |
| Between rows (i.e., between varieties of fertilizers) | 18 | $(4-1)=3$ | $18/3=6$ | $6/1=6$ | $F(3,6)=4.76$ |
| Residual or error | 6 | $(3-1)\times(4-1)=6$ | $6/6=1$ | | |
| Total | 32 | $(3\times4)-1=11$ | | | |

From the said ANOVA table, we find that differences concerning varieties of seeds are insignificant at 5% level as the calculated *F*-ratio of 4 is less than the table value of 5.14, but the variety differences concerning fertilizers are significant as the calculated *F*-ratio of 6 is more than its table value of 4.76.